

Deep Learning for NLP

BERT

Measuring Performance



Learning goals

- Task-specific metrics
- Metrics for text output
- How to evaluate retrieval
- Subtle aspects of evaluation

TASK-SPECIFIC EVALUATION: CLASSIFICATION

(Binary) Document-level Classification

- Accuracy
- Recall / Precision
- F1-Score
- ROC-Curve / AUC
- ...

(Multi-Class) Document-level classification

- Micro- / Macro-averaged F1
- Class-specific accuracies
- ...

TASK-SPECIFIC EVALUATION: RANKING (1)

Ranking:

- Use Cases:
 - Retrieval systems output a ranking of relevant documents
 - Search engines return top k results
 - ...
- Ranking evaluation metrics
 - Mean average precision (MAP)
 - Mean reciprocal rank (MRR)
 - Precision@ k / Recall@ k
 - ...

TASK-SPECIFIC EVALUATION: RANKING (2)

MAP:

$$AP@k = \frac{1}{r} \sum_{i=1}^k precision@i \cdot R_i$$

r = number of relevant items

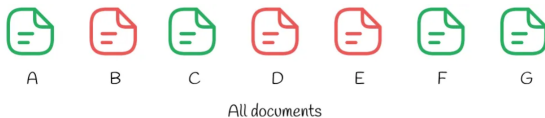
$$R_i = \begin{cases} 1, & \text{if document } i \text{ is relevant} \\ 0, & \text{if document } i \text{ is not relevant} \end{cases}$$

 k | $precision@k$

► Source: towardsdatascience

TASK-SPECIFIC EVALUATION: RANKING (3)

Example MAP:



► Source: [towardsdatascience](https://towardsdatascience.com/evaluating-ranking-models-part-3-map-precision-recall-auc-4000000000000000)

TASK-SPECIFIC EVALUATION: RANKING (4)

Example MAP:



Retrieved documents for query Q_1

$$AP@5 = \frac{1}{3} \left(\frac{1}{1} + \frac{2}{3} + \frac{3}{4} \right) = 0.81$$

k	precision@k
1	$\frac{1}{1}$
2	$\frac{1}{2}$
3	$\frac{2}{3}$
4	$\frac{3}{4}$
5	$\frac{3}{5}$

► Source: towardsdatascience

TASK-SPECIFIC EVALUATION: RANKING (5)

Example MAP:



B



F



E



C



A

Retrieved documents for query Q_2

$$AP@5 = \frac{1}{3} \left(\frac{1}{2} + \frac{2}{4} + \frac{3}{5} \right) = 0.53$$

k	precision@k
1	$\frac{0}{1}$
2	$\frac{1}{2}$
3	$\frac{1}{3}$
4	$\frac{2}{4}$
5	$\frac{3}{5}$

► Source: towardsdatascience

TASK-SPECIFIC EVALUATION: RANKING (6)

MAP:

$$MAP = \frac{1}{|Q|} \sum_{q \in Q} AP_q@k, \text{ with } Q = \text{set of queries}$$



$$MAP@5 = \frac{1}{3} (0.81 + 0.68 + 0.50) = 0.66$$

► Source: towardsdatascience

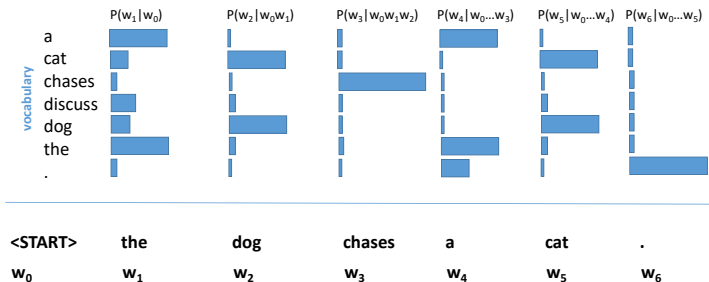
TASK-SPECIFIC EVALUATION: RANKING (7)

MRR:

$$RR = \frac{1}{\text{rank of first relevant item}}$$

$$MRR = \frac{1}{|Q|} \sum_{q \in Q} RR_q, \text{ with } Q = \text{set of queries}$$

ARLMS



- **Question:** How can we measure “performance” on this task?

ARLM PERPLEXITY (1)

- Natural choice for evaluating generated text
- Well-defined for ARLMs; trickier for MLMs
- Intuitively: *Amount of the model's surprisal when confronted with a sequence* (higher value means higher “surprisal”)
- More technically: Measure of uncertainty of a probabilistic model
- Example: Perplexity of a fair k -sided die (uniform distribution) is k

ARLM PERPLEXITY (2)

- Autoregressive factorization of the sequence:

Hugging Face is a startup based in New York City and Paris
p(word)

► Source: huggingface

- Log-probability of i-th token given context: $\log(p_{\theta}(w_i | w_{<i}))$
more “certain” model $\rightarrow$ higher log-probability
- Aggregate¹ over the whole sequence:

$$PPL = \exp \left(-\frac{1}{t} \sum_{i=1}^t \log(p_{\theta}(w_i | w_{<i})) \right)$$

¹The choice of the log's base is basically arbitrary.

ARLM PERPLEXITY (3)

- *Lower bound* is 1, i.e. the model predicts *every* token correctly (with a certainty of 100%)
- **Question:** Is this really desirable?
- *Upper bound* is $|V|$, i.e. the model only provides a random guess for every token with a probability of $\frac{1}{|V|}$
- Selection of state-of-the-art perplexities:
 - **1B Word Benchmark** ▶ Chelba et al., 2013 ($|V| = 800k$)
PPL = 21,8 ▶ Dai et al., 2019
 - **WikiText-103** ▶ Merity et al., 2016 ($|V| = 270k$)
PPL = 10,8 ▶ Shueybi et al., 2019
 - **Penn Treebank** ▶ Marcus et al., 1994 ($|V| = 10k$)
PPL = 20,5 ▶ Brown et al., 2020

ARLM PERPLEXITY (4)

- *Problem:* Fixed context size of models (e.g. 1024 for GPT-2)

Hugging Face is a startup based in New York City and Paris

$p(\text{word}|\text{context})$

► Source: huggingface

- *Possible solution:* Sliding window strategy
- Close approximation to “true” autoregressive decomposition
- *Drawback:* Computationally expensive (individual forward pass for each token)

ARLM PERPLEXITY (5)

Question:

Is a language model with lower perplexity always a “better” model?

Some (Dis)Advantages of perplexity:

- **Pro:** Straight-forward applicable
- **Pro:** Does not depend on “external” labels
- **Con:** Not clear what value can be considered “good”
- **Con:** Corpus-/language-specific measure

EVALUATING GENERATED TEXT (1)

- **Question:** How else can we evaluate the quality of generated text?
- *Use cases:*
 - Machine translation
 - Question answering (extractive or abstractive)
 - Dialogue generation
 - Text summarization
 - Image Captioning
 - Code generation

EVALUATING GENERATED TEXT (2)

Machine Translation

- Metrics based on N-gram-overlap
 - BLEU (cf. Chap. 3.1) ▶ Papineni et al., 2002
 - ROUGE ▶ Lin, 2004
 - METEOR ▶ Banerjee and Lavie, 2005
- Metrics based pre-trained (neural) models
 - BertScore ▶ Zhang et al., 2019
 - BLEURT ▶ Sellam et al., 2020
 - COMET ▶ Rei et al., 2020

EVALUATING GENERATED TEXT (3)

- BLEU, ROUGE, METEOR:
 - "Bilingual evaluation understudy", originally for machine translation
 - Measures overlap of words, and longer word sequences of different sizes
 - BLEU: unreliable per-example, but correlates with human judgments on an aggregate level ▶ Reiter, 2018

EVALUATING GENERATED TEXT (3)

- BERTScore
 - Based on pre-trained BERT model
 - Also able to consider contextual similarities
 - Correlates better with human judgements
- BLEURT
 - Better per-example quality estimation by fine-tuning on examples with judgements
 - But: potential bias if tested systems are very different than training examples

EVALUATING GENERATED TEXT (4)

Question Answering / Summarization / Dialogues

- Aspects to consider
 - Factual correctness
 - Fluency
 - Stylistic aspects
 - Engagement
 - ...
- Human evaluation?! (cf. next slide)

HUMAN EVALUATION

Pros

- Possible where simply using metrics fails (esp. generated texts)
- Probably gold standard (given trained human evaluator)
- *Strong* learning signal for models (cf. RLHF, chapter 9)
- ...

Cons

- Manual, tedious work
- Costs and working conditions
- Subjectivity
- Ambiguity
- ...